

Track 7: Cloud & Server Technologies (DevOps)

June 2026

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MR ROKESH TECHNOLOGY — Training & Internship Programs

Skill path: Linux → Git → Docker → Kubernetes → CI/CD → Monitoring → IaC

Related: [Master Syllabus](#) | [Duration Framework](#) | [Track 8 — Cloud Platforms](#) (AWS/Azure/GCP services)

Track Overview

Who this is for: Aspiring DevOps engineers, SREs, and backend developers managing infrastructure.

Prerequisites: Basic command-line familiarity helpful; Linux taught from Week 1.

Tools & lab environment: Linux VM (Ubuntu), Git/GitHub, Docker, Docker Compose, Kubernetes (Minikube/kind), GitHub Actions, Jenkins, Terraform, Ansible, Prometheus, Grafana, Loki, ArgoCD.

Distinction from Track 8: Track 7 focuses on **how to build, deploy, and operate** systems (DevOps toolchain). Track 8 focuses on **cloud provider services** (AWS EC2, S3, Lambda, etc.).

Certification targets (6–9 month): CKA (Kubernetes) or Docker DCA prep (optional)

Outcomes by Duration

Duration	Hours	Job Readiness	Key Deliverables
2 Weeks	40–50	DevOps awareness	Deploy containerized app live
1 Month	80–100	Junior DevOps basics	App deployed on Minikube
3 Months	250–300	Internship-ready	Microservices on K8s capstone
6 Months	500–600	DevOps engineer entry	Production observable system
9 Months	750–900	SRE/GitOps ready	GitOps platform + self-healing architecture

Weekly Rhythm (All Durations)

Mon–Thu infrastructure labs | Fri automation milestone | Sat pipeline review | Sun runbook

documentation

2-Week Crash Course

Mini Project: Deploy containerized app live | **Level:** L1

Days	Topics	Lab	Self-Check
D1-2	Linux fundamentals, SSH, file permissions, shell scripting	Linux admin tasks	Write bash script with loops
D3-4	Git/GitHub, branching, PR workflow	Collaborative repo project	Resolve merge conflict
D5-6	Docker: images, containers, Dockerfile, Compose	Containerize a web app	Explain image vs container
D7-8	CI/CD with GitHub Actions	Auto-build + test pipeline	YAML pipeline triggers on push
D9-10	Basic monitoring, deploy to cloud	Deploy containerized app live	Health check endpoint verified

1-Month Foundation

Capstone: Deploy app on Minikube | **Level:** L2

Week	Topics	Project
W1	Linux admin + shell scripting	Automated backup script
W2	Git deep dive + GitHub Actions CI	CI pipeline for Node app
W3	Docker + Docker Compose multi-service	3-tier app in containers
W4	Kubernetes basics (pods, services, deployments)	Deploy app on Minikube

W4 acceptance criteria: - ☐ Deployment, Service, and Ingress YAML applied - ☐ Rolling update demonstrated - ☐ CI pipeline builds and pushes Docker image

3-Month Job Ready

Capstone: Containerized microservices on K8s | Level: L3

Month	Topics	Projects
M1	Linux, Git, shell scripting, networking	Automation scripts
M2	Docker, Compose, registry, CI/CD (GitHub Actions/Jenkins)	CI/CD pipeline
M3	Kubernetes, Helm basics, Prometheus/Grafana intro	Containerized microservices on K8s

6-Month Professional

DevOps Core Course 18-lab track aligned — [GitHub](#)

Month	Topics	Labs
M1	Linux, Git internals, CI/CD foundations	Labs 1–3
M2	Docker, Terraform, Ansible	Labs 4–6
M3	Logging (Loki), Monitoring (Prometheus/Grafana)	Labs 7–8
M4	Kubernetes, Helm, secrets management	Labs 9–12
M5	GitOps (ArgoCD), progressive delivery	Labs 13–14
M6	StatefulSets, cluster monitoring, capstone	Production-grade observable system

M6 Capstone acceptance criteria (L4): - [] 3+ microservices with Docker Compose and K8s manifests - [] Prometheus metrics + Grafana dashboard - [] Centralized logging with Loki or ELK - [] GitOps deployment via ArgoCD - [] Runbook for common failure scenarios

9-Month Advanced

Capstones: GitOps platform + Self-healing multi-service architecture | **Level:** L5

Additional modules: Full IaC (Terraform + Pulumi), service mesh (Istio intro), SRE practices (SLI/SLO), DevSecOps (Trivy, OWASP ZAP in pipeline), multi-cloud deployment.

Assessment Rubric (Track-Specific)

Component	Weight	Criteria
Weekly Quiz	20%	Linux, Docker, K8s, CI/CD concepts
Lab Completion	40%	Scripts run; pipelines green; clusters healthy
Project	30%	Architecture, automation, observability
Portfolio	10%	GitHub with IaC repos, pipeline YAML, runbooks

Recommended Resources

- [DevOps Core Course \(18 Labs\)](#)
- [Docker Documentation](#)
- [Kubernetes Documentation](#)

Appendix: Mentor Notes

Common pitfalls: Running containers as root; secrets in Git; no health checks; skipping rollback testing.

Hardware: 16 GB RAM for Minikube; WSL2 on Windows recommended.

Cross-track: [Track 8 — Cloud Platforms](#) for EKS/ECS deployment; [Track 1 — Full Stack](#) for app being deployed.